

Cost Minimization Tutorial

Quick Solutions

↳ In the tutorial, make sure to go slow: draw the isoquants, the iso-cost lines, etc.

Problem 1

(a) $\frac{1}{2}$ unit of water

(b) (4, 2) units of light and water respectively

(c) $h = l$, $h = 2w$

$$\Rightarrow l^c(w_1, w_2, h) = h$$

$$w^c(w_1, w_2, h) = h/2$$

$$(d) C(w_1, w_2, h) = w_1 h + w_2 \frac{h}{2}$$

Problem 2

(a) 1 hour

(b) $1/2$ bag

(c) It would be cheaper to talk

(d) the smallest between

$$\frac{w_f}{2} h \text{ OR } w_t h$$

$$\Rightarrow C(w_f, w_t, h) = \min \left\{ \frac{w_f}{2}, w_t \right\} h$$

$$(e) t^c(w_f, w_t, h) = \begin{cases} h & \text{if } w_t < w_f/2 \\ 0 & \text{if } w_t > w_f/2 \end{cases}$$

Problem 3

$$(a) f(tx, tg) = (tg)(tx)^2$$

$$= t^2 g x^2 = t^3 f(x, g)$$

$$> t f(x, g)$$

$\Rightarrow$ increasing returns

$$(b) 1440 = 10 x^2$$

$$\Rightarrow 144 = x^2 \Rightarrow x = 12$$

Costs are: $12w_x + 10w_g$

$$(c) y = 10x^2 \Rightarrow x = \sqrt{\frac{y}{10}}$$

$$\Rightarrow C(w_x, w_g, y, g_0 = 10) =$$

$$w_x \sqrt{\frac{y}{10}} + w_g 10$$

$$(d) AT(w_x, w_g, y) = \frac{w_x \sqrt{y/10} + 10w_g}{y}$$

$$AVC(w_x, w_g, y) = \frac{w_x \sqrt{y/10}}{y}$$

$$AFC(w_x, w_g, y) = 10w_g/y$$

$$MC(w_x, w_g, y) = \frac{w_x}{\sqrt{10}} \cdot \frac{1}{2\sqrt{y}} = \frac{w_x}{2\sqrt{10}y}$$